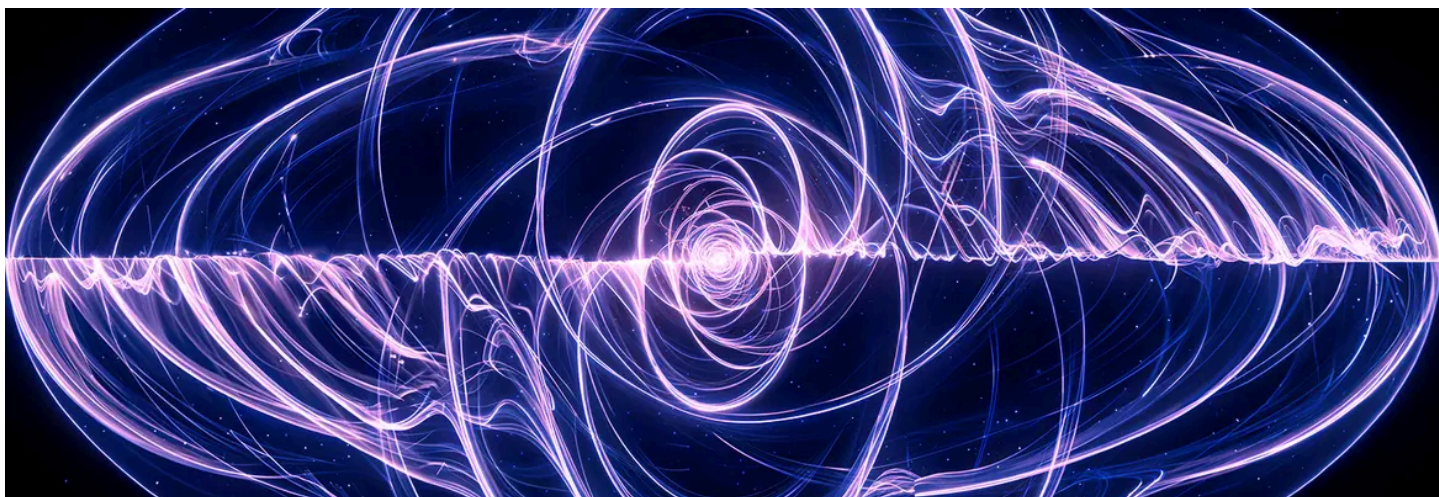


news

[Home](#) [Philosophy](#) [Science](#) [Politics](#) [Arts](#) [Issue Archive](#)

Schrödinger's blunder created the quantum measurement problem

Quantum reality has both waves and particles



16th September 2026

 [cite](#)

Antony Valentini | Academic Visitor at Imperial College London, and author of *Beyond the Quantum: A Quest for the Origin and Hidden Meaning of Quantum Mechanics* (Oxford University Press, 2025).

2,732 words

Read time: approx. 14 mins

*Quantum mechanics is seen as challenging the idea that science provides objective descriptions of reality outside of our minds. The “measurement problem” seems to suggest that the act of observation creates the outcome that is observed. But physicist **Antony Valentini**, whose *Beyond the Quantum* was hailed by Lee Smolin as “the best book written this century” on quantum mechanics, argues that this picture rests on an enormous blunder by Erwin Schrödinger. After Louis de Broglie predicted the wave-like trajectories of particles, Schrödinger found the equation describing this. But whereas de Broglie envisaged particles riding on waves, Schrödinger kept the waves and threw away the particles. That single decision birthed the measurement problem—and it was completely unnecessary. Resurrect de Broglie’s original particle-based theory, Valentini argues, and we can dispel quantum mystery.*

Get the
big ideas straight to
your inbox weekly

Enter Email Address

[Sign me up](#)

Related Posts:

Reality and the quantum

Across the sciences, realism reigns. Astrophysicists observe galaxies that are billions of years old; geologists study the Earth's deep history; evolutionary biologists sequence ancient DNA from fossils. It is understood that galaxies and the Earth formed long before the rise of humans, and that now-extinct species lived and died millions of years ago. Only in the peculiar context of quantum physics do we find claims that physical reality somehow depends on the presence of human observers, even though humans did not evolve until long after the formation of galaxies, stars, and planets, and arrived quite late on the scene of life on Earth. Are such claims grounded in special evidence? Is there perhaps some remarkable laboratory phenomenon that demonstrates the power of human observers to “create reality”?

According to many physics textbooks, as well as popular accounts, the iconic two-slit experiment (Figure 1) demonstrates that there is **no objective reality of moving particles**. It is often argued that the appearance of a localised particle at the backstop is a random event triggered by the conscious intervention of “the observer” (causing the wave-like state to “collapse” to a particle-like state). **God plays dice**. Particles do not exist until they are measured. And similarly for cats, the Moon, and the wider universe.

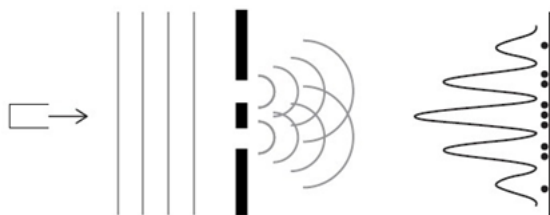
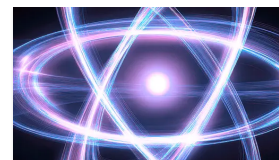


Figure 1. The two-slit experiment.

Such claims are belied by “pilot-wave theory,” first developed by French physicist Louis de Broglie in the 1920s, then revived by American physicist David Bohm in the 1950s. This is **a theory of particles in motion, guided by waves**, as illustrated in Figure 2. As a crude analogy, think of a floating bottle carried along by an ocean wave. The wave hits the barrier and passes through the slits, so two waves emerge on the far side. A single particle carried by the wave moves from source to backstop, along a definite path or trajectory, and through one hole only. Over many repetitions, the particle trajectories bunch into the supposedly inexplicable “interference pattern” predicted by quantum mechanics (Figure 3)—provided the positions of the incoming particles are distributed in a way that matches the incoming wave (technically, its squared-magnitude, the famous “Born probability rule”). In similar fashion, the theory gives the same predictions as quantum mechanics for other experiments too. Even so, the theory has been mostly swept under the rug by physicists and historians.



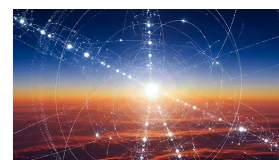
The musical structure behind quantum physics



Chemistry is not explained by quantum mechanics



The quantum world reveals reality is made of relations, not objects



The universe is fine-tuned for technology, not just life

Related Videos:



Time's great mystery

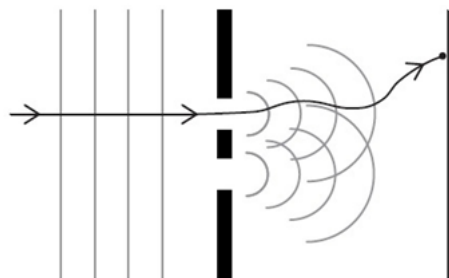
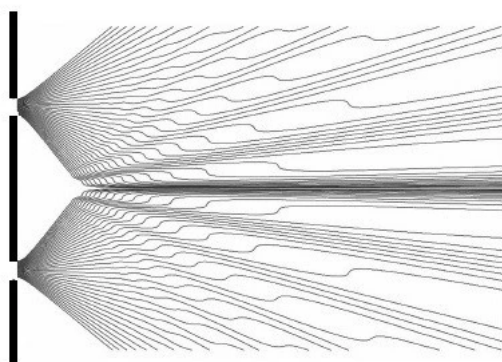
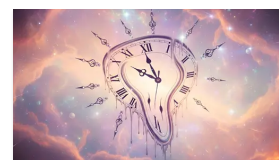


Figure 2. Pilot-wave theory for one particle.



Our future
theories of the
universe



Physics has
misunderstood
time



The trouble with
string theory

Want to continue reading?

Get unlimited access to insights from the world's leading
thinkers.

Browse our [subscription plans](#) and subscribe to read more.

Start Free Trial

Already a subscriber? [Log in](#)

LATEST RELEASES

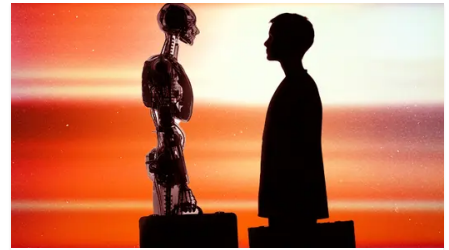




Gender equality and the freedom trap



Activists don't need to win in order to achieve their aims, with Sophie Scott-Brown



AI taking your job isn't the only problem, it's dividing workers too

Join the conversation

Sign in to post comments or **join now (only takes a moment)**.



© The Institute of Art and Ideas



Get iai email updates

I would like to receive updates from the Institute of Art and Ideas.

Click Here

No spam ever. You can unsubscribe at any time with just one click.